MULTI-CENTER PROTOTYPE ENHANCED MULTIMODAL LOW-RANK PROMPTING FOR PHASE-RESOLVED PARTIAL DISCHARGE RECOGNITION

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ABSTRACT

Phase-Resolved Partial Discharge (PRPD) recognition aims to identify insulation defects using partial-discharge patterns of power equipment. Existing methods suffer from significant intra-class morphological diversity, inter-class feature overlap, as well as the efficiency-accuracy trade-off. To tackle these issues, we propose MCP-MMLoP, a dual-module framework consisting of an MMLoP module and a multi-center prototype (MCP) module. The MMLoP module leverages multimodal low-rank prompting for parameter-efficient model adaptation, yet it suffers from limited representation capacity due to low-rank subspace constraints, which hinders handling the challenge that patterns within the same discharge category exhibit high morphological variability and mitigates insufficiently the issue that features from distinct discharge types overlap with one another. The MCP module maintains multiple sub-prototypes per discharge class to explicitly characterize diverse intra-class morphologies and supply complementary discriminative cues, alleviating the challenges arising from intra-class variations and inter-class similarity. Experiments on a PRPD dataset with six discharge classes demonstrate that MCP-MMLoP introduces only 0.0092M additional trainable parameters compared with the MMLoP baseline and achieves state-of-the-art performance. The results verify that our framework improves discriminative ability for complex PRPD samples while preserving the parameter-efficiency benefits of low-rank prompt learning.

1 INTRODUCTION

Partial Discharge (PD) is an important precursor of insulation degradation in power equipment (Bartnikas, 2002; Judd et al., 2005). PRPD patterns contain rich information of discharge types and are widely adopted for insulation condition assessment and fault diagnosis (Shahsavarian et al., 2021; Florkowski, 2020; Stone, 2005). However, automated PRPD recognition faces inherent difficulties: patterns within the same discharge category exhibit high morphological variability, features from distinct discharge types overlap with one another, and models struggle to balance recognition accuracy and computational cost, as illustrated in Figure 1.

Existing PRPD recognition methods fall into two main categories: unimodal PRPD image recognition (Abubakar & Zachariades, 2024; Jamil et al., 2026)and multimodal PRPD recognition (Chouaibi et al., 2025; Jia et al., 2026a). Unimodal PRPD image recognition takes only PRPD patterns as input and performs identification via handcrafted features or convolutional networks, yet it lacks physical semantics and exhibits limited modeling capacity for complex samples (Wu et al., 2023; Yan et al., 2025). Multimodal PRPD recognition fuses multi-source information such as patterns and text, and leverages vision-language models (VLMs) to incorporate domain priors. Nevertheless, it typically incurs substantial parameter overhead and struggles to balance efficiency and accuracy (Abubakar & Zachariades, 2024; Hu et al., 2023).

Multimodal methods can be further divided into traditional multimodal fusion methods and prompt learning-based vision-language adaptation methods. Traditional multimodal methods integrate discharge waveforms, statistical features and pattern information through feature concatenation and weighted fusion. However, they rely on manually designed fusion rules, resulting in limited gener-

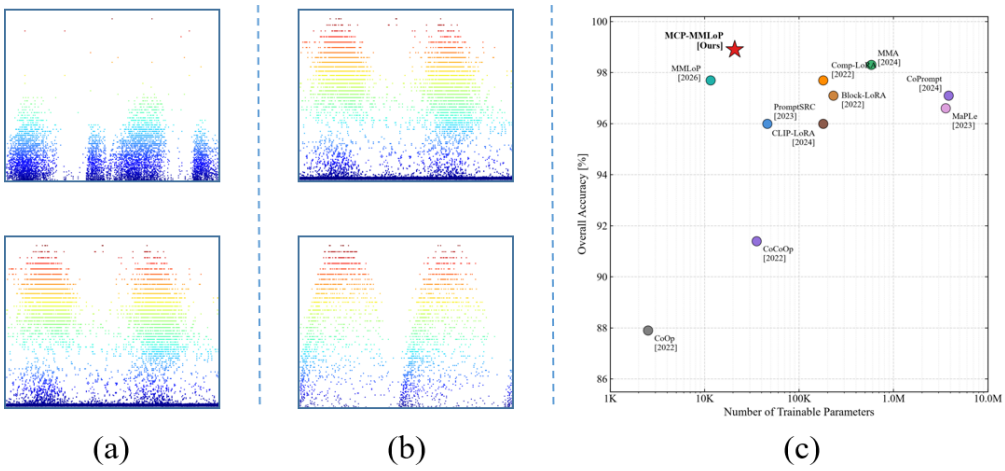


Figure 1: Three difficulties automated PRPD recognition faced: (a) Patterns within the same discharge category exhibit high morphological variability. (b) Features from distinct discharge types overlap with one another. (c) Models struggle to balance recognition accuracy and computational cost.

alization capability and poor adaptability to few-shot scenarios. Recent advances in VLMs, especially CLIP (Radford et al., 2021), have provided a promising paradigm for few-shot recognition. Prompt learning freezes the backbone of pre-trained VLMs and optimizes only a small set of task-specific parameters to achieve efficient model adaptation (Jia et al., 2022). Representative methods including CoOp (Zhou et al., 2022b), CoCoOp (Zhou et al., 2022a), MaPLe (Khattak et al., 2023a), and PromptSRC (Khattak et al., 2023b) have significantly improved the flexibility and generalization performance of prompt-based adaptation. Therefore, our work adopts prompt learning-based vision-language adaptation for PRPD recognition.

Existing prompt learning-based vision-language adaptation methods have demonstrated promising performance. For instance, MaPLe (Khattak et al., 2023a) achieves precise cross-modal alignment by inserting learnable prompts into both vision and text encoders, yet it suffers from heavy parameter and computational costs, limiting its lightweight adaptation capability. Instead of encoder prompt tuning, MMA (Yang et al., 2024) performs multimodal fusion via feature stacking and projection. Despite its concise architecture, it lacks sufficient cross-modal interaction modeling and yields weakly discriminative features, making it ill-suited for complex PRPD samples. MMLoP (Ghiasvand et al., 2026) adopts low-rank prompt parameterization and a shared cross-modal up-projection matrix, constraining prompt parameters within a low-dimensional subspace to drastically reduce parameter counts while preserving cross-modal interaction modeling. Nevertheless, none of these approaches simultaneously tackle three core challenges: patterns within the same discharge category exhibit high morphological variability, features from distinct discharge types overlap with one another, and models struggle to balance recognition accuracy and computational cost.

To address these limitations, this work proposes the Multi-Center Prototype Enhanced Multimodal Low-Rank Prompting (MCP-MMLoP) framework, which consists of an MMLoP module and an MCP module. The MMLoP module leverages multimodal low-rank prompting for parameter-efficient model adaptation, yet it suffers from limited representation capacity due to low-rank subspace constraints, which hinders modeling of complex intra-class discharge morphologies and mitigates inter-class feature overlap insufficiently. Thus, we introduce the MCP module, which maintains multiple sub-prototypes per discharge class to explicitly characterize diverse intra-class morphologies and supply complementary discriminative cues. We further adopt entropy-weighted exponential moving average (EMA) and prototype-space regularization to stabilize prototype updates and enlarge inter-class prototype separation. Output from the two modules are weighted and fused for final recognition. We construct a PRPD dataset containing 1694 patterns across six discharge categories. Our main contributions are summarized as follows:

- We propose the MCP module, which assigns multiple sub-prototypes to each discharge class and characterizes diverse pattern morphologies within the same discharge category via maximum similarity aggregation. It explicitly models intra-class multimodal distributions to mitigate intra-class variations and inter-class similarity in PRPD recognition.

- To resolve the efficiency-accuracy trade-off, we introduce the MMLoP module (Ghiasvand et al., 2026). This module constructs low-rank parameterized prompts and a shared cross-modal up-projection structure. By constraining prompt parameter distribution within a low-dimensional subspace, it drastically reduces parameter overhead while stably preserving cross-modal interaction modeling capacity.

- We build a PRPD dataset covering six typical discharge morphologies, including Suspension Discharge, Tip Discharge, Surface Discharge, Air Gap discharge, Metal Discharge, and noisy normal samples, corresponding to common insulation defect scenarios in power equipment. Extensive experiments demonstrate that MCP-MMLoP achieves state-of-the-art performance on this dataset.

2 Related Work

2.1 PRPD Pattern Recognition

PRPD pattern analysis is widely used for insulation fault diagnosis in power equipment. Early studies relied on statistical, fractal, and conventional machine-learning features (Sahoo et al., 2005; Basharan, 2018), while subsequent works adopted CNN-based models for PRPD or discharge-signal recognition (Peng et al., 2019; Liu et al., 2021; Sun et al., 2021). More recent studies explore image-based matching, data augmentation, and few-shot metric learning (Jia et al., 2026b; Wang et al., 2023; Li et al., 2025).

2.2 Prompt Learning-Based Vision-Language Adaptation Methods

VLMs provide a promising foundation for few-shot recognition by transferring knowledge from large-scale image-text pretraining. CLIP (Radford et al., 2021) enables zero-shot recognition through aligned visual and textual representations. Building upon CLIP, CoOp (Zhou et al., 2022b) and CoCoOp (Zhou et al., 2022a) introduce learnable and instance-conditioned prompts, respectively, while MaPLe (Khattak et al., 2023a), PromptSRC (Khattak et al., 2023b), MMA (Yang et al., 2024), and CoPrompt (Roy & Etemad, 2024) further improve multimodal prompt adaptation. However, multimodal deep prompting can introduce considerable trainable parameters, motivating parameter-efficient alternatives including visual prompt tuning, adapters, and low-rank adaptation (Jia et al., 2022; Houlsby et al., 2019; Hu et al., 2022; Zanella & Ben Ayed, 2024). MMLoP (Ghiasvand et al., 2026) addresses this issue through low-rank prompt parameterization with a shared up-projection matrix, reducing trainable parameters to approximately 11.5 K while maintaining competitive performance.

2.3 Prototype Learning

Prototypical Networks (Snell et al., 2017) represent each class with a single prototype, yet a single center may be insufficient for classes where patterns within the same discharge category exhibit high morphological variability. This issue is closely related to few-shot diagnosis under limited or ambiguous supervision (Feng et al., 2020; Li et al., 2023; Wang et al., 2023). Subsequent methods have explored mixture-based or MCP to capture heterogeneous class structures. Infinite Mixture Prototypes (Allen et al., 2019) model classes with multiple mixture components, while MCP employs multiple sub-prototypes with EMA, intra-class compactness, and inter-class separation to improve discrimination under limited supervision.

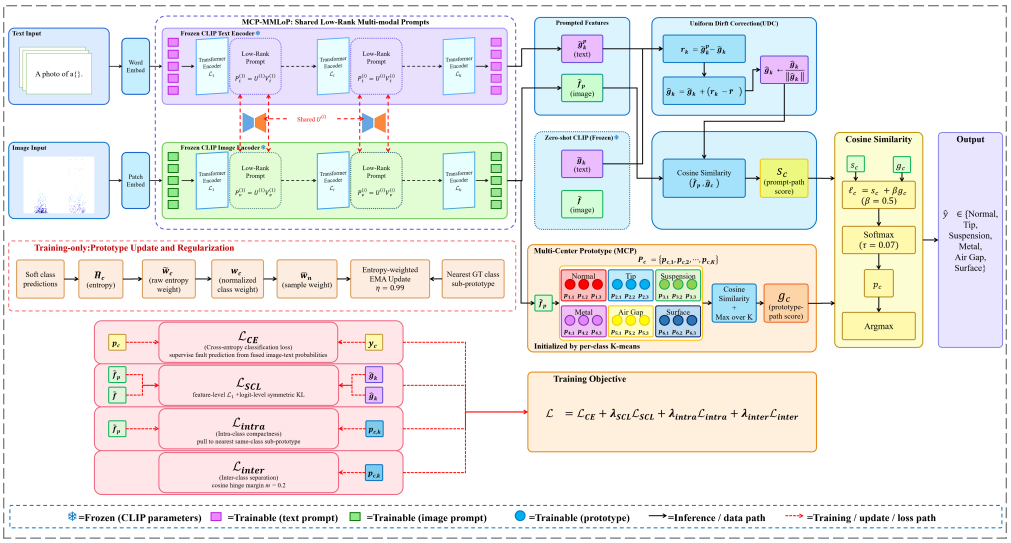


Figure 2: Overall framework of the proposed MCP-MMLoP.

3 METHOD

3.1 PROBLEM DEFINITION AND OVERALL FRAMEWORK

Let the training set be $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N}$, where x_n denotes a PRPD pattern and $y_n \in 1, \ldots, C$ denotes the discharge-type label ($C = 6$: Tip, Metal, Air Gap, Suspension, Surface, and Normal). MCP-MMLoP adopts a frozen CLIP dual-encoder as its backbone. Neither the image encoder $f(\cdot)$ nor the text encoder $g(\cdot)$ is updated, and only two types of lightweight parameters are optimized: the low-rank prompt factors $\{\mathbf{U}^{(l)}, \mathbf{V}_v^{(l)}, \mathbf{V}_t^{(l)}\}_{l=1}^{L}$ and the MCP $\{\mathbf{P}_c\}_{c=1}^{C}$.

The overall framework is illustrated in Figure 2. Given an input pattern x, the image encoder with visual prompts produces the prompted feature $\tilde{\mathbf{f}}_p$. The text encoder with textual prompts produces the prompted text feature $\tilde{\mathbf{g}}_k^p$, which is subsequently corrected by Uniform Drift Correction (UDC) to obtain $\hat{\mathbf{g}}_k$. The two paths compute class scores in parallel: the prompt-path score is $s_c = \mathrm{sim}(\tilde{\mathbf{f}}_p, \hat{\mathbf{g}}_c)$, while the prototype-path score is $g_c = \max_k \mathrm{sim}(\tilde{\mathbf{f}}_p, \mathbf{p}_{c,k})$. The final score $\ell_c = s_c + \beta g_c$ is converted into class probabilities through temperature-scaled softmax.

3.2 LOW-RANK MULTIMODAL PROMPTING

Following MMLoP (Ghiasvand et al., 2026), each layer-wise prompt matrix is parameterized as the product of two low-rank factors. Let the visual and textual prompts at the l-th layer be $\mathbf{P}_v^{(l)} \in \mathbb{R}^{V \times d_v}$ and $\mathbf{P}_t^{(l)} \in \mathbb{R}^{T \times d_t}$, respectively. Then,

$$\mathbf{P}_v^{(l)} = \mathbf{U}^{(l)} \mathbf{V}_v^{(l)}, \qquad \mathbf{P}_t^{(l)} = \mathbf{U}^{(l)} \mathbf{V}_t^{(l)}, \tag{1}$$

where $\mathbf{U}^{(l)} \in \mathbb{R}^{T \times r}$ is the shared up-projection matrix shared across the visual and textual modalities, with T set to the visual prompt length; the visual branch reuses it after dimensional adaptation. $\mathbf{V}_v^{(l)} \in \mathbb{R}^{r \times d_v}$ and $\mathbf{V}_t^{(l)} \in \mathbb{R}^{r \times d_t}$ are the down-projection matrices, and the rank satisfies $r \ll \min(d_v, d_t)$. The inputs to the prompted Transformer layers are given by

$$\tilde{\mathbf{X}}_p^{(l)} = \{\mathbf{P}_v^{(l)}, \mathbf{e}_{\text{cls}}^{(l)}, \mathbf{e}_1^{(l)}, \ldots, \mathbf{e}_M^{(l)}\}, \qquad \tilde{\mathbf{Y}}_p^{(l)} = \{\mathbf{t}_{\text{SOS}}^{(l)}, \mathbf{P}_t^{(l)}, \mathbf{t}_1^{(l)}, \ldots, \mathbf{t}_N^{(l)}, \mathbf{c}_k^{(l)}, \mathbf{t}_{\text{EOS}}^{(l)}\}. \tag{2}$$

During training, the CLIP backbone θ_{CLIP} is completely frozen, and only the low-rank factors are optimized.

3.3 CLASSIFICATION OUTPUT OF THE PROMPT PATH AND UDC

The prompted image feature is denoted as $\tilde{\mathbf{f}}_p$, the prompted text feature of class k as $\tilde{\mathbf{g}}_k^p$, and the zero-shot features of the frozen CLIP as $\tilde{\mathbf{f}}$ and $\tilde{\mathbf{g}}_k$. Prompt tuning introduces systematic drift across all class embeddings, i.e., a "base-class bias," thereby degrading generalization. To address this issue, UDC is adopted. The prompted text feature is decomposed into a zero-shot anchor and a residual, $\mathbf{r}_k = \tilde{\mathbf{g}}_k^p - \tilde{\mathbf{g}}_k$, and the mean residual shared across classes, $\bar{\mathbf{r}} = \frac{1}{C}\sum_k \mathbf{r}_k$, is subtracted:

$$\hat{\mathbf{g}}_k = \tilde{\mathbf{g}}_k + (\mathbf{r}_k - \bar{\mathbf{r}}), \qquad \hat{\mathbf{g}}_k \leftarrow \hat{\mathbf{g}}_k / \|\hat{\mathbf{g}}_k\|. \tag{3}$$

The corrected features retain only the class-discriminative residuals. The score of the prompt path for class c is computed using cosine similarity:

$$s_c = \mathrm{sim}(\tilde{\mathbf{f}}_p, \hat{\mathbf{g}}_c) = \frac{\tilde{\mathbf{f}}_p \cdot \hat{\mathbf{g}}_c}{\|\tilde{\mathbf{f}}_p\| \, \|\hat{\mathbf{g}}_c\|}. \tag{4}$$

3.4 MCP MODULE

Accordingly, we introduce the concept of sub-prototypes to characterize a specific morphology within each discharge class.

$$\mathbf{P}_c = \{\mathbf{p}_{c,1}, \mathbf{p}_{c,2}, \ldots, \mathbf{p}_{c,K}\}, \qquad \mathbf{p}_{c,k} \in \mathbb{R}^D, \tag{5}$$

Where D denotes the feature dimension, consistent with $\tilde{\mathbf{f}}_p$. Sub-prototypes are initialized with the K-means cluster centers of per-class training features, so they correspond to distinct pattern morphologies from the start of training. The cosine similarity between sample $\tilde{\mathbf{f}}_p$ and sub-prototype $\mathbf{p}_{c,k}$ is defined as

$$\mathrm{sim}(\tilde{\mathbf{f}}_p, \mathbf{p}_{c,k}) = \frac{\tilde{\mathbf{f}}_p \cdot \mathbf{p}_{c,k}}{\|\tilde{\mathbf{f}}_p\| \, \|\mathbf{p}_{c,k}\|}. \tag{6}$$

The prototype-path score for class c is defined as the maximum similarity among its K sub-prototypes (soft assignment, where a sample only needs to match its most similar morphology):

$$g_c(\tilde{\mathbf{f}}_p) = \max_{k \in \{1,\ldots,K\}} \mathrm{sim}(\tilde{\mathbf{f}}_p, \mathbf{p}_{c,k}). \tag{7}$$

Max aggregation enables each sub-prototype to independently characterize one pattern morphology, avoiding the averaging of different morphologies into a single center. This constitutes the fundamental distinction between multi-center and single-center prototypes.

3.5 EMA PROTOTYPE UPDATE

Prototypes need stable evolution during training. Direct prototype update via batch gradients is vulnerable to perturbations from confusing samples such as Tip-Metal boundary samples. We thus adopt the EMA update(Zhang et al., 2026). The batch-level class marginal distribution and information entropy are computed first. Let $\boldsymbol{q}_n$ denote the soft label of sample n. The marginal probability of class c in a batch is:

$$\bar{p}_c = \frac{1}{B}\sum_{n=1}^{B} q_{n,c}, \tag{8}$$

and its information entropy $\bar{H}_c = -\bar{p}_c \log \bar{p}_c$ measures the dispersion of the current predictions for this class. A larger entropy indicates greater uncertainty in class discrimination, and thus its contribution to prototype updates should be reduced:

$$\tilde{w}_c = \frac{1}{\bar{H}_c}, \qquad w_c = \frac{\tilde{w}_c}{\frac{1}{C}\sum_{j=1}^{C} \tilde{w}_j}, \tag{9}$$

where the normalization ensures $\frac{1}{C}\sum_c w_c = 1$, preventing systematic drift in the update magnitude. The sample-level weight is obtained by aggregating the class weights according to the soft label: $\bar{w}_n = \sum_{c=1}^{C} q_{n,c}\, w_c$.

For sample n, its most similar sub-prototype in the ground-truth class is $k_n^* = \arg\max_k \mathrm{sim}(\tilde{\mathbf{f}}_{p,n}, \mathbf{p}_{y_n,k})$. The corresponding sub-prototype is updated via EMA with weight $\bar{w}_n$:

$$\mathbf{p}_{y_n,k_n^*}^{t+1} = \eta\, \mathbf{p}_{y_n,k_n^*}^{t} + (1-\eta)\, \bar{w}_n\, \tilde{\mathbf{f}}_{p,n}, \tag{10}$$

where the momentum $\eta \in [0, 1)$ is set to 0.99 in the final model, following the sensitivity analysis in Table 5. The contributions of samples from high-entropy (uncertain) classes are automatically attenuated, preventing confused samples from significantly perturbing the prototype locations.

3.6 Prototype Space Regularization

To mitigate intra-class dispersion and inter-class overlap, we adopt two lightweight regularizers:

Intra-class compactness: samples are encouraged to move toward the most similar sub-prototype of their corresponding class,

$$\mathcal{L}_{\text{intra}} = \frac{1}{B}\sum_{n=1}^{B} \left(1 - g_{y_n}(\tilde{\mathbf{f}}_{p,n})\right). \tag{11}$$

This loss only acts on the nearest sub-prototype, allowing different morphologies within the same class to form their own compact clusters rather than forcing the entire class into a single cluster.

Inter-class separation: a cosine hinge constraint is imposed on arbitrary pairs of sub-prototypes from different classes, penalizing their similarity when it exceeds the margin m:

$$\mathcal{L}_{\text{inter}} = \frac{1}{|\mathcal{S}|} \sum_{(c,k)\neq(c',k')} \max\left(0,\ \mathrm{sim}(\mathbf{p}_{c,k}, \mathbf{p}_{c',k'}) - m\right), \tag{12}$$

where $\mathcal{S}$ denotes the set of all cross-class sub-prototype pairs, and $m = 0.2$ is the margin. The hinge formulation drives the loss to zero once the required margin is satisfied, thereby avoiding unnecessary over-separation.

3.7 Dual-Path Fusion and Overall Objective

The scores from the two paths are weighted and fused:

$$\ell_c = s_c + \beta\, g_c(\tilde{\mathbf{f}}_p), \tag{13}$$

where β is the fusion coefficient, set to 0.5 by default. The class probabilities are obtained using temperature-scaled softmax:

$$p_c = \frac{\exp(\ell_c/\tau)}{\sum_{c'=1}^{C} \exp(\ell_{c'}/\tau)}, \tag{14}$$

where the temperature τ follows the CLIP convention and is set to 0.07. The prompt path provides cross-modal semantic discrimination, while the prototype path provides multiple morphological centers within each class; the two paths are complementary.

The overall training objective is defined as

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{SCL}}\mathcal{L}_{\mathrm{SCL}} + \lambda_{\mathrm{intra}}\mathcal{L}_{\mathrm{intra}} + \lambda_{\mathrm{inter}}\mathcal{L}_{\mathrm{inter}}, \tag{15}$$

where $\mathcal{L}_{\mathrm{CE}}$ denotes the cross-entropy loss computed from the fused probabilities, and $\mathcal{L}_{\mathrm{SCL}}$ denotes the self-regulated consistency loss (Ghiasvand et al., 2026) from MMLoP, which consists of a feature-level L1 constraint and a logit-level symmetric KL constraint:

$$\begin{aligned}\mathcal{L}_{\mathrm{SCL}} = \lambda_1\|\hat{\mathbf{g}}_k - \tilde{\mathbf{g}}_k\|_1 + \lambda_2\|\tilde{\mathbf{f}}_p - \tilde{\mathbf{f}}\|_1 \\ + \tfrac{1}{2}D_{\mathrm{KL}}\big(\mathrm{sim}(\tilde{\mathbf{f}}_p, \hat{\mathbf{g}}_k) \,\|\, \mathrm{sim}(\tilde{\mathbf{f}}, \tilde{\mathbf{g}}_k)\big) \\ + \tfrac{1}{2}D_{\mathrm{KL}}\big(\mathrm{sim}(\tilde{\mathbf{f}}, \tilde{\mathbf{g}}_k) \,\|\, \mathrm{sim}(\tilde{\mathbf{f}}_p, \hat{\mathbf{g}}_k)\big),\end{aligned} \tag{16}$$

3.8 Training Algorithm

The training procedure of MCP-MMLoP is summarized in Algorithm 1.

Algorithm 1 MCP-MMLoP Training and Inference

Require: Training set $\mathcal{D}$, frozen CLIP encoders f, g, rank r, prompt depth $L = 9$, number of sub-prototypes K, loss weights $\lambda_{\mathrm{SCL}}, \lambda_{\mathrm{intra}}, \lambda_{\mathrm{inter}}$, fusion coefficient β.

Ensure: Low-rank factors $\{\mathbf{U}^{(l)}, \mathbf{V}_v^{(l)}, \mathbf{V}_t^{(l)}\}$ and multi-center prototypes $\{\mathbf{P}_c\}$.

Initialize $\mathbf{U}^{(l)}, \mathbf{V}_v^{(l)}, \mathbf{V}_t^{(l)} \sim \mathcal{N}(0, 0.05)$ for $l = 1, \ldots, L$.
Initialize $\{\mathbf{p}_{c,k}\}$ by applying K-means to the training features of each class.
for each training iteration **do**
  Sample a mini-batch $\{(\mathbf{x}_n, y_n)\}$.
  Construct low-rank prompts using Eq. (1) and obtain $\tilde{\mathbf{f}}_p$ and $\tilde{\mathbf{g}}_{p,k}$.
  Apply UDC correction using Eq. (3) to obtain $\hat{\mathbf{g}}_k$.
  Compute $s_c = \mathrm{sim}(\tilde{\mathbf{f}}_p, \hat{\mathbf{g}}_c)$.
  Compute $g_c = \max_k \mathrm{sim}(\tilde{\mathbf{f}}_p, \mathbf{p}_{c,k})$ and $\ell_c = s_c + \beta g_c$.
  Compute $\mathcal{L}_{\mathrm{CE}}$ and $\mathcal{L}_{\mathrm{SCL}}$.
  Compute entropy and sample weights using Eqs. (8)–(9).
  Compute $\mathcal{L}_{\mathrm{intra}}$ and $\mathcal{L}_{\mathrm{inter}}$ using Eqs. (11)–(12).
  Update the low-rank factors by minimizing $\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{SCL}}\mathcal{L}_{\mathrm{SCL}} + \lambda_{\mathrm{intra}}\mathcal{L}_{\mathrm{intra}} + \lambda_{\mathrm{inter}}\mathcal{L}_{\mathrm{inter}}$.
  Update prototypes using EMA according to Eq. (10).
end for
Inference: $\hat{y} = \arg\max_c(s_c + \beta g_c)$.

4 Experiments

4.1 Dataset and Experimental Protocol

Dataset. We evaluate MCP-MMLoP on a laboratory-based PRPD dataset containing 1,694 patterns from six discharge classes: Tip, Suspension, Surface, Air Gap, Metal, and Normal. Each pattern is represented as a PRPD scatter plot with phase angle on the horizontal axis and discharge amplitude on the vertical axis. The dataset is divided into training, validation, and test sets with 1353, 167, and 174 samples, respectively, as summarized in Table 3 of Appendix A.

Experimental Protocol. The dataset is stratified by class and randomly split at a ratio of 8:1:1. To reduce the effect of random partitioning, all experiments are repeated with five random seeds, and the mean performance is reported.

4.2 Implementation Details

We use CLIP ViT-B/16 with frozen image and text encoders. Inputs are resized to 224×224 and normalized using CLIP statistics. Due to the physical semantics of PRPD patterns, horizontal and vertical flipping are disabled; mild brightness jitter, Gaussian noise ($\sigma \leq 0.01$), and small-angle rotation ($\leq 5^\circ$) are applied. Following the default MMLoP configuration, we use $T = V = 4$ prompt tokens, inject prompts into the first 9 Transformer layers, and adopt rank-1 ($r = 1$) factorization with a shared up-projection matrix. We set $\lambda_1 = 25$, $\lambda_2 = 10$, $K = 3$, $\tau = 0.07$, $m = 0.2$, and $\eta = 0.99$. AdamW is used with learning rates of 1×10^{-3} and 1×10^{-4} for prompt and prototype parameters, respectively, cosine annealing, and weight decay of 5×10^{-4}. The batch size is 32, and training lasts 100 epochs with early stopping (patience 10). Experiments are conducted in PyTorch with mixed precision on a 24 GB NVIDIA RTX 3090 GPU.

4.3 Evaluation Metrics

Due to class imbalance in our dataset, we report overall accuracy(OA), macro-P, macro-R, macro-F1, Cohen's Kappa, and per-class recall, alongside confusion matrices for error analysis. Statistical significance relative to the strongest baseline is assessed via a paired t-test ($\alpha = 0.05$), with corresponding p-values reported.

4.4 Comparison Methods

Table 1: Full-supervision performance comparison. Mean results over 5 random seeds are reported in %.

Method	Params.	Tip	Suspension	Surface	Air Gap	Metal	Normal	OA	Macro-F1
Zero-shot CLIP (Radford et al., 2021)	0	89.7	0.0	0.0	0.0	0.0	0.0	20.1	5.7
CoOp (Zhou et al., 2022b)	2.5 K	97.4	88.9	84.6	80.0	79.5	96.3	87.9	88.4
CoCoOp (Zhou et al., 2022a)	35.4 K	100.0	83.3	80.8	92.0	89.7	96.3	91.4	90.6
MaPLe (Khattak et al., 2023a)	3.56 M	94.9	100.0	100.0	96.0	92.3	100.0	96.6	97.2
PromptSRC (Khattak et al., 2023b)	46.08 K	100.0	100.0	96.2	92.0	89.7	100.0	96.0	96.2
MMA (Yang et al., 2024)	0.58 M	94.9	100.0	100.0	96.0	100.0	100.0	98.3	98.6
CoPrompt (Roy & Etemad, 2024)	3.82 M	97.4	100.0	100.0	92.0	94.9	100.0	97.1	97.5
CLIP-LoRA (Zanella & Ben Ayed, 2024)	0.18 M	94.9	100.0	96.2	92.0	94.9	100.0	96.0	96.5
Block-LoRA (Hu et al., 2022)	0.23 M	94.9	100.0	96.2	96.0	97.4	100.0	97.1	97.5
Comp-LoRA (Hu et al., 2022)	0.18 M	94.9	100.0	96.2	100.0	97.4	100.0	97.7	98.1
MMLoP (Ghiasvand et al., 2026)	11.56 K	100.0	100.0	96.2	100.0	92.3	100.0	97.7	98.1
MCP-MMLoP(Ours)	20.8 K	100.0	100.0	96.2	100.0	97.4	100.0	98.9	99.0

All methods use the same CLIP ViT-B/16 backbone, data splits, and data augmentation. We compare against CLIP zero-shot recognition, prompt-learning methods including CoOp (Zhou et al., 2022b), CoCoOp (Zhou et al., 2022a), MaPLe (Khattak et al., 2023a), PromptSRC (Khattak et al., 2023b), MMA, and CoPrompt, as well as parameter-efficient adaptation methods including CLIP-LoRA (Zanella & Ben Ayed, 2024), Block-LoRA, and Comp-LoRA (Hu et al., 2022). MMLoP (Ghiasvand et al., 2026) serves as the primary low-rank prompting baseline. For fair comparison, all methods use the text template "a photo of {}" when applicable. Trainable parameter counts are reported together with recognition performance in Table 1.

Table 1 summarizes the comparison results. We highlight three main observations.

1. **Prompt learning substantially improves zero-shot recognition.** Zero-shot CLIP achieves only 20.1% OA, with zero recall on five of the six discharge classes. CoOp improves OA to 87.9%, demonstrating the importance of adapting CLIP to the PRPD domain.
2. **MMLoP provides strong parameter efficiency.** With only 11.56 K trainable parameters, MMLoP achieves 97.7% OA, outperforming MaPLe (96.6%) while using only about 0.3% of its 3.56 M parameters. This demonstrates the effectiveness of low-rank prompt adaptation for parameter-efficient PRPD recognition.
3. **MCP-MMLoP achieves the best performance.** MCP-MMLoP obtains 98.9% OA and 99.0% macro-F1 with 20.8 K trainable parameters, improving over MMLoP by 1.2 and 0.9

percentage points, respectively. The largest gain is observed for Metal Discharge, whose recall increases from 92.3% to 97.4%, indicating that multi-center prototypes help distinguish morphologically similar discharge patterns. The improvement over the second-best baseline, MMA (98.3% OA), is statistically significant ($p < 0.05$).

Figure 3 presents the few-shot results. As the number of training samples increases from 4 to 32 shots per class, both OA and macro-F1 improve across methods. MCP-MMLoP consistently achieves the best performance, with a larger advantage over MMLoP in the 4- and 8-shot settings. This suggests that multi-center prototypes provide complementary structural information when supervision is limited, while the prompt branch benefits from increased supervision at higher shot levels.

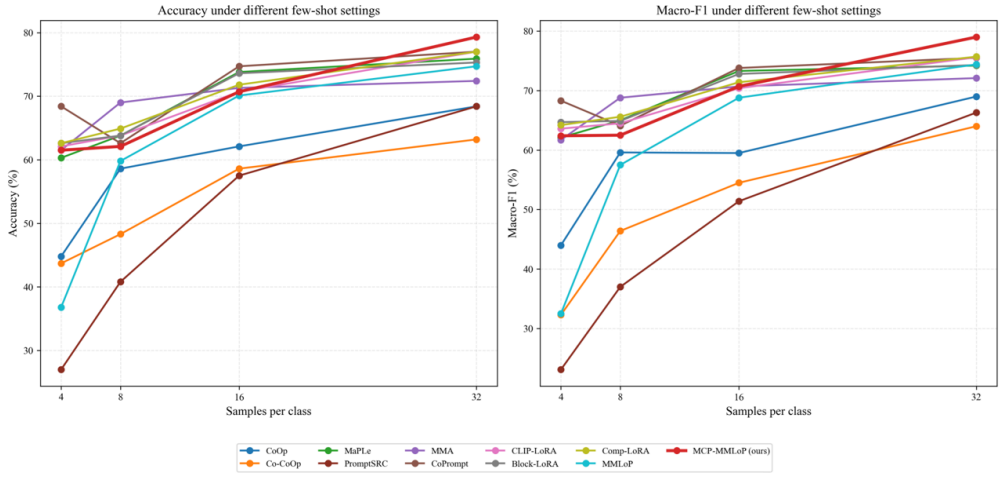


Figure 3: Comparison of OA and macro-F1 under different few-shot settings.

4.5 Ablation Study

Table 2: Ablation study of the proposed MCP-MMLoP (%, mean over 5 random seeds).

Module	Tip	Suspension	Surface	Air Gap	Metal	Normal	OA	macro-F1
Baseline (MMLoP (Ghiasvand et al., 2026))	100.0	100.0	96.2	100.0	92.3	100.0	97.7	98.1
w/ $\mathcal{L}_{\text{intra}}$	100.0	100.0	92.3	96.0	87.2	100.0	95.4	95.5
w/ $\mathcal{L}_{\text{inter}}$	100.0	100.0	92.3	92.0	89.7	100.0	95.4	95.5
MCP-MMLoP	**100.0**	**100.0**	**96.2**	**100.0**	**97.4**	**100.0**	**98.9**	**99.0**

To isolate the contributions of the multi-center prototype module, we evaluate four variants using MMLoP as the baseline: (i) MMLoP without the prototype path and regularization terms; (ii) MMLoP with only $\mathcal{L}_{\text{intra}}$; (iii) MMLoP with only $\mathcal{L}_{\text{inter}}$; and (iv) the full MCP-MMLoP with the prototype path and both regularization terms, as shown in Table 2 .

The results show that adding either regularization term alone reduces OA from 97.7% to 95.4%, suggesting that geometric constraints without prototype-path guidance can interfere with prompt learning. In contrast, jointly introducing the multi-center prototype path with $\mathcal{L}_{\text{intra}}$ and $\mathcal{L}_{\text{inter}}$ improves OA to 98.9%, demonstratparting the complementarity of intra-class attraction and inter-class separation. The largest gain is observed for Metal Discharge, whose recall increases from 92.3% to 97.4%, indicating that multi-center prototypes better capture the morphological diversity between Metal and Surface Discharges.

4.6 Computational Efficiency Analysis

The multi-center prototype module introduces merely $C \times K \times D = 6 \times 3 \times 512 \approx 9.2$ K additional parameters and requires only one extra matrix multiplication during inference. Compared with

MMLoP, it achieves a 1.2% OA improvement with a less than twofold parameter increase, yielding state-of-the-art recognition performance. These results verify that the proposed module delivers prominent classification gains and improved interpretability via prototype-based representations, with negligible parameter and computational overhead.

5 Conclusion

In this work, we proposed MCP-MMLoP, a dual-module framework that complements low-rank multimodal prompt learning with multi-center prototype modeling for few-shot PRPD pattern recognition. Future work will investigate domain generalization and weakly supervised learning to improve robustness across devices and operating conditions.

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A Supplementary Experimental Details

A.1 Dataset Statistics and Representative Patterns

Table 3 reports the class-wise split used in all experiments. Stratified splitting is adopted to maintain the original class distribution across training, validation and test partitions and mitigate the influence of class imbalance. Figure 4 provides representative PRPD patterns from the six discharge classes..

Table 3: PRPD dataset statistics.

Class	Typical physical cause	Training	Validation	Test
Tip Discharge	Local field concentration at metal protrusion	307	38	39
Metal Discharge	PD induced by free moving metal particles	307	38	39
Surface Discharge	Creeping discharge along oil-pressboard interface	196	24	26
Suspension Discharge	Charge accumulation on floating metal parts	134	16	18
Air Gap Discharge	Gas-void breakdown inside solid insulation	200	25	25
Normal	Background noise without insulation defect	209	26	27
Total		1353	167	174

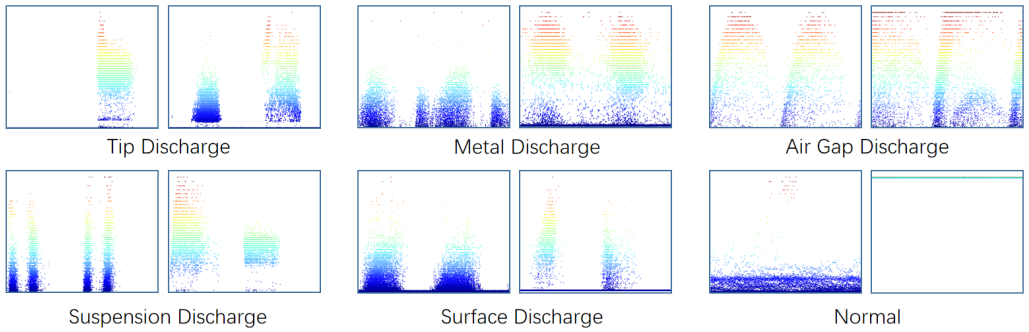


Figure 4: Examples of PRPD patterns for six discharge classes, with two samples shown per class. Notice that patterns within the same discharge category exhibit high morphological variability, alongside the fact that features from distinct discharge types overlap with one another, leading to recognition ambiguity. For instance, patterns from Metal Discharge and Air Gap Discharge exhibit visually similar distributions, posing challenges for discriminative learning.

A.2 Implementation Details and Hyperparameters

For reproducibility, key implementation hyper-parameters are summarized in Table 4. Under the parameter-efficient adaptation paradigm, the full CLIP vision-language backbone (both image and text encoders) is frozen throughout training to preserve general vision-text alignment knowledge pre-trained on large-scale external datasets. Only lightweight low-rank prompt factors and multi-center prototype parameters are updated via back-propagation. This setup restricts learnable parameters to a small model subset, alleviating the accuracy-computation trade-off while leveraging the strong pre-trained representations from CLIP.

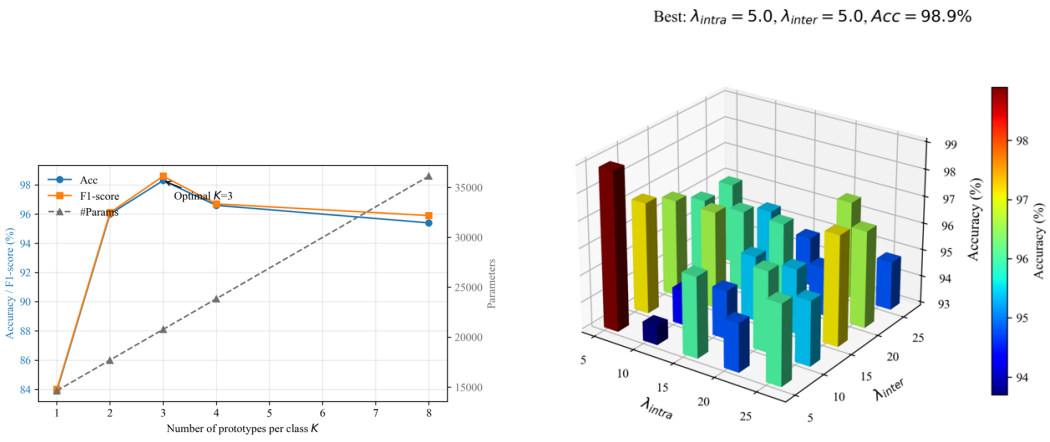
B Additional Analysis

B.1 Hyperparameter Sensitivity

The number of sub-prototypes and the prototype regularization weights are analyzed in Figures 5. Modifying these hyper-parameters alters prototype representation capacity and the geometric constraints applied on the learned feature space. The selected setting $K = 3$, $\lambda_{\text{intra}} = 5.0$, and $\lambda_{\text{inter}} = 5.0$ provides the best trade-off between recognition performance and model complexity.

Table 4: Summary of hyperparameters.

Module	Hyperparameter	Value
Low-rank prompting	Rank r / prompt length $T = V$ / number of layers L	1 / 4 / 9
Multi-center prototypes	Number of sub-prototypes K / temperature τ / margin m / EMA momentum η	3 / 0.07 / 0.2 / 0.99
Loss weights	λ_1 / λ_2 / λ_{intra} / λ_{inter} / β	25 / 10 / 5.0 / 5.0 / 0.5
Training	Learning rate (prompt / prototype) / batch size / epochs	1e−3 / 1e−4 / 32 / 100



(a) (b)

Figure 5: Joint caption for combined figure. (a) Sensitivity analysis of the number of sub-prototypes K.; (b) Joint sensitivity analysis of regularization weights λ_{intra} and λ_{inter}.

Increasing K enables finer modeling of heterogeneous sample distributions, yet it adds extra trainable parameters; overly large regularization coefficients, on the other hand, may constrain prompt learning and degrade overall accuracy. Table 5 further shows that $\eta = 0.99$ gives the best EMA performance. A sufficiently large momentum smooths prototype updates and reduces instability caused by noisy or confusing training samples.

Table 5: Effect of different EMA momentum coefficients η (%, mean over 5 random seeds).

η	OA	macro-F1
0.1	96.0	96.5
0.3	94.8	95.1
0.5	95.4	95.9
0.7	96.6	96.8
0.9	96.6	97.0
0.95	97.7	98.1
0.99	98.9	99.0

B.2 Training Convergence

Figure 6 shows stable convergence: validation accuracy rises rapidly in the early epochs and stabilizes later, while the validation loss decreases and converges with only minor fluctuations. This observation indicates that the joint optimization of low-rank prompt factors and multi-center prototypes works well under our entropy-weighted EMA update rule. No severe oscillation or divergence is observed throughout the training process. After rapid improvement in the initial training phase, both metrics gradually settle, which demonstrates that the proposed dual-module framework can reach a well-behaved optimum without additional optimization tricks.

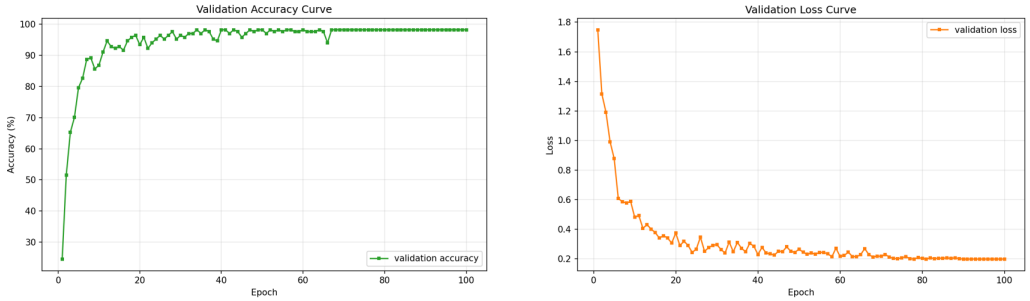


Figure 6: Validation accuracy and loss curves over training epochs.

B.3 Additional Visualization and Error Analysis

Figure 7 shows that the remaining errors are concentrated among these visually similar classes. Such misclassifications correspond to the intrinsically ambiguous samples of the PRPD task, where fault patterns exhibit high visual resemblance across different discharge categories. Figure 8 compares the feature distributions of 12 methods via t-SNE projection. MCP-MMLoP yields clearer class separation in the latent feature space, particularly for the morphologically similar Metal and Surface Discharge classes, which are challenging to distinguish for competing approaches.

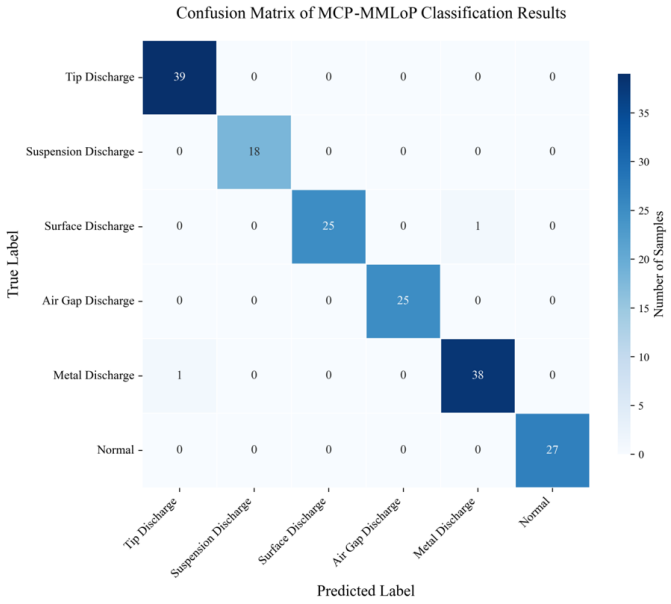
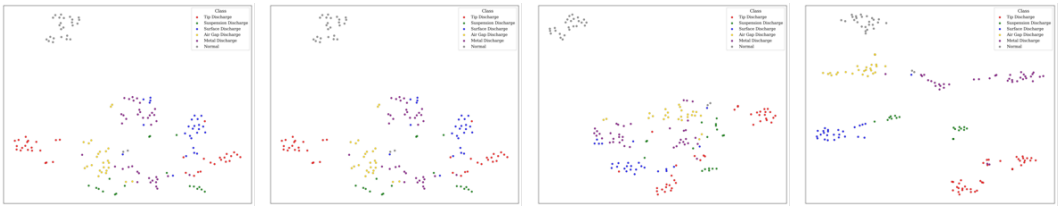
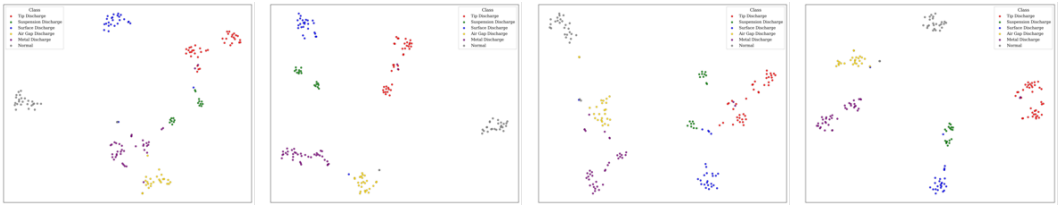


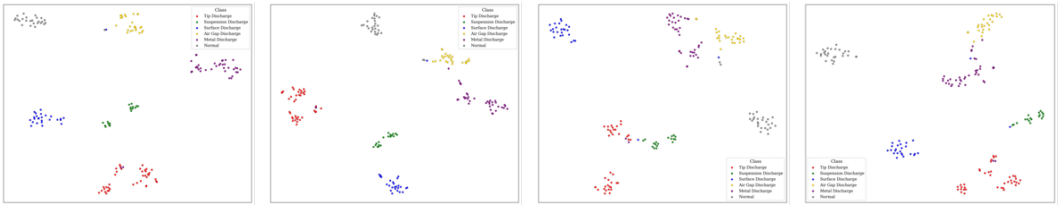
Figure 7: Confusion matrix of classification results obtained by MCP-MMLoP on the test set.



(a) Zero-shot CLIP 0.1352 (b) CoOp 0.3352 (c) Co-CoOp 0.3647 (d) MaPLe 0.4689



(e) PromptSRC 0.4829 (f) MMA 0.4893 (g) CoPrompt 0.3993 (h) CLIP-LoRA 0.5343



(i) Block-LoRA 0.5417 (j) Comp-LoRA 0.5327 (k) MMLoP 0.5351 (l) MCP-MMLoP (ours) 0.6761

Figure 8: t-SNE visualization of test-set features extracted by different methods.